

Course: B. Tech Branch: Electronics Engineering & Allied

Subject Code & Name: BTETOE604C/BTEXOE604C Computer Network / Computer Network & Cloud Computing

Semester: VI

Max Marks: 60

Date: 26/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 is compulsory.
3. Attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Objective type questions. (Compulsory Question)		12
1 Which type of topology is best suited for large businesses which must carefully control and coordinate the operation of distributed branch outlets? a. Ring b. Local area c. Hierarchical d. Star	Understand	1
2 What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences? a. Optical fiber b. Coaxial cable c. Unshielded twisted pair d. Microwave	Understand	1
3 A collection of hyperlinked documents on the internet forms the ?? a. WWW b. E-mail system c. Mailing list d. Hypertext markup language	Remember	1
4 The term FTP stands for? a. File transfer program b. File transmission protocol c. File transfer protocol d. File transfer protection	Remember	1
5 Which of the following address belongs class A? a. 121.12.12.248 b. 130.12.12.248 c. 128.12.12.248 d. 129.12.12.248	Understand	1
6 Which layer of the TCP / IP stack corresponds to the OSI model transport layer? a. Host to host b. Application c. Internet d. Network Access	Understand	1
7 In specific, if the systems use separate protocols, which one of the following devices is used to link two systems? a. Repeater b. Gateway c. Bridge d. Hub	Understand	1
8 What is the maximum efficiency of pure aloha at $G = 1/2$?	Apply	1

	a. 1.89	b. 17.99	c. 18.999	d. 18.4		
9	Which of the following protocols is the bit-oriented protocol?				Remember	1
	a. SSL	b. http	c. HDLC	d. All of the these		
10	What is the size of the destination port in the UDP protocol?				Remember	1
	a. 8 bits	b. 16 bits	c. 20 bits	d. 32 bits		
11	What is the size of the sender window in the Go Back N (ARQ) protocol?				Understand	1
	a. 0	b. 1	c. 10	d. n		
12	In which of the following switching methods, the message is divided into small packets?				Understand	1
	a. Message switching	b. Packet switching	c. Virtual switching	d. None of the these		
Q.2	Solve the following.					12
A)	Explain the OSI reference model with neat diagram				Understand	6
B)	What are the different types of networks? Explain in detail.				Understand	6
Q.3	Solve the following.					12
A)	Explain HDLC protocol in detail				Understand	6
B)	What are the types of class full addressing? Find the class of the following IP addresses.				Apply	6
	a) 208.34.54.12	b) 238.34.2.1	c) 14.34.2.8			
Q.4	Solve Any Two of the following.					12
A)	Write Short Note on				Understand	6
	a) IPV4 Addressing					
	b) IPV6 Addressing					
B)	Compare the different sliding window protocols				Analyze	6
C)	Explain in brief about different framing methods				Understand	6
Q.5	Solve Any Two of the following.					12
A)	Explain different types of switching techniques				Understand	6
B)	Explain the following networking devices in detail.				Understand	6
	a) Switch					
	b) Router					
	c) Bridge					
C)	Explain the various transmission media in detail.				Understand	6

Q.6 Solve Any Two of the following.

12

A) Write Short Note on following:

Understand

6

a. Domain Name Space (DNS)

b. File Transfer Protocol (FTP)

B) Explain the TCP/IP reference model with neat diagram

Understand

6

C) What is network topology? Explain the different network topologies.

Understand

6

***** End *****

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